Transfer task: defend and revise

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Defend the prediction with evidence and make a visible, justified revision.

SUCCESS CRITERIA - I can...

1. Graph the central prediction and coordinate bounds.
2. Defend a claim using calculations and model limits.
3. Document a specific revision with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A student says a 3 kg cart under 15 N net force accelerates at 45 m/s². Identify the error, correct it and name a check you can now do independently.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Use your Day 21 data and consistent units. Defend the endpoint and assumptions. A bounded range is not a probability.

Worked reasoning: A cart model predicts 12 m after 4 s; the observation is 10±0.5 m. Since 12 is outside 9.5-10.5 m, the model needs checking. The mismatch alone does not identify the cause.

Your turn: Mark the observed interval and prediction on a number line. Name one model assumption to recheck.

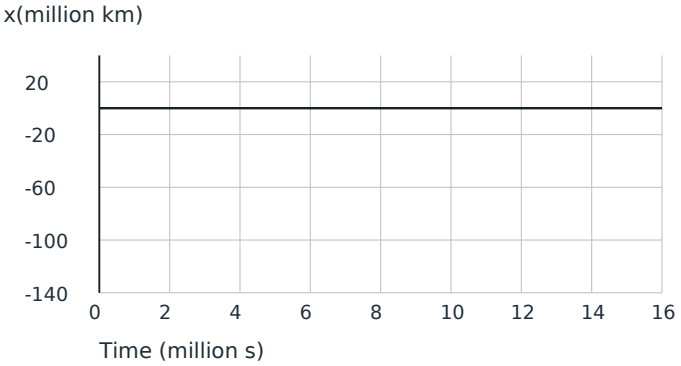


Position (m); right is positive

Marzano check - circle: ① ② ③

03 Show the prediction

Using your Day 21 CPA work, draw an x-versus-time graph. Connect the central start and finish. Then add lower and upper prediction lines. Label time in millions of seconds and x in millions of km.



Marzano check - circle: ① ② ③

04 Write a public briefing

Write 4-5 sentences: central position; range; assumption; one excluded physical effect; what you can and cannot claim about Earth impact.

Explain to your audience	Evidence from your work
Predicted position	____
Range and meaning	____
Assumption / omitted effect	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Defend and revise

Ask a partner to check one calculation and one graph label. Record one question, then revise a specific step or reason why it should stay. If no partner is available, perform and label a self-review.

Support: I changed (or kept)____ because ____.

Step checked	Question or concern	Change or reason to keep
____	____	____

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 6 kg sign hangs still from one vertical cord. Use g=10 N/kg. Name both forces, their sources, magnitudes and directions.
Draw the forces on the chosen object and label their sources.



Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

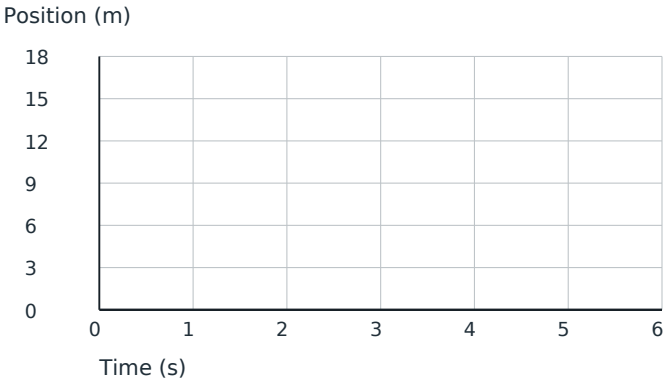
Point to one piece of work that meets each success criterion. Name one claim you can now support and one question about curving paths that requires Unit 2.

Criterion	Where your work shows it
Science / model	____
Reasoning	____
Communication / transfer	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Revise a constant-speed model from a 0 m start to match the central observation of 10 m after 4 s. Propose a new observation that could test the revision.



Marzano check - circle: 1 2 3